

Foundation (Jalapeno)

Q1. Factor the expression: $x^2 + 5x + 4x + 20$

- (A) $x(x + 5) + 4(x + 5)$
- (B) $x(x + 4) + 5(x + 4)$
- (C) $x(x + 1) + 4(x + 5)$
- (D) $x(x + 2) + 5(x + 4)$
- (E) $x(x + 3) + 4(x + 5)$

Q2. Factor the expression: $3x^2 + 12x + 4x + 16$

- (A) $3x(x + 4) + 4(x + 4)$
- (B) $3x(x + 2) + 4(x + 8)$
- (C) $3x(x + 1) + 4(x + 12)$
- (D) $3x(x + 3) + 4(x + 4)$
- (E) $3x(x + 5) + 4(x + 2)$

Q3. Factor the expression: $2x^2 + 5x + 3x + 7$

- (A) $2x(x + 3) + 7(x + 1)$
- (B) $2x(x + 2) + 5(x + 3)$
- (C) $2x(x + 5) + 3(x + 1)$
- (D) $2x(x + 1) + 7(x + 3)$
- (E) $2x(x + 4) + 5(x + 2)$

Q4. Factor the expression: $x^2 + 2x + 6x + 12$

- (A) $x(x + 2) + 6(x + 2)$
- (B) $x(x + 6) + 2(x + 6)$
- (C) $x(x + 1) + 6(x + 1)$
- (D) $x(x + 5) + 2(x + 5)$
- (E) $x(x + 3) + 6(x + 4)$

Q5. Factor the expression: $x^2 + 4x + 3x + 12$

- (A) $x(x + 3) + 4(x + 3)$
- (B) $x(x + 4) + 3(x + 4)$
- (C) $x(x + 2) + 6(x + 4)$
- (D) $x(x + 6) + 2(x + 3)$
- (E) $x(x + 1) + 12(x + 3)$

Q6. Factor the expression: $2x^2 + 7x + 2x + 3$

- (A) $2x(x + 2) + 7(x + 1)$
- (B) $2x(x + 1) + 3(x + 7)$
- (C) $2x(x + 7) + 3(x + 1)$
- (D) $2x(x + 2) + 3(x + 2)$
- (E) $2x(x + 3) + 2(x + 3)$

Q7. Factor the expression: $x^2 + 3x + 2x + 6$

- (A) $x(x + 2) + 3(x + 2)$
- (B) $x(x + 3) + 2(x + 3)$
- (C) $x(x + 4) + 1(x + 3)$
- (D) $x(x + 6) + 3(x + 1)$
- (E) $x(x + 1) + 6(x + 3)$

Q8. Factor the expression: $3x^2 + 2x + 9x + 6$

- (A) $3x(x + 2) + 6(x + 1)$
- (B) $3x(x + 1) + 2(x + 9)$
- (C) $3x(x + 3) + 6(x + 2)$
- (D) $3x(x + 6) + 2(x + 3)$
- (E) $3x(x + 2) + 3(x + 2)$

Open Ended Questions (FRQ)

Q9. Factor the expression: $x^2 + 5x + 4x + 20$, then verify by multiplication.

Q10. Factor the expression: $2x^2 + 7x + 4x + 14$, then verify by multiplication.

Chef's Tips

- Tip 1: Make sure to check the factoring by multiplying the factors together.
- Tip 2: Emphasize the importance of grouping when factoring expressions.
- Tip 3: Have students work in pairs to verify each other's work.